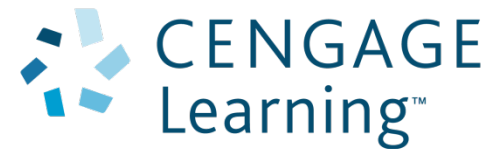


Chapter 9

Linear Momentum and Collisions



Outline for W9,D2

Ch 8) Cons of energy with friction

Ch 8) Power, $P=dE/dt$, $P=F \cdot v$

Ch 9) Momentum, $\mathbf{p}=m\mathbf{v}$

Conservation of momentum – 1D examples

Relation to force

Homework

Ch. 9 P. 1,2,4-6,8,9,18,21,23,28,29,37,38,47,54,55 Due Mon

Notes: “NEW STUFF”:

Practice quizzes, exam-like Qs, Ch.8 PPT, Ch 9 PPT,
old notes with examples, etc.

Including friction in energy problems (copied from chapter7.ppt)

A good general equation for including both friction and other external forces is:

$$\Delta E_m = W_{\text{ext}} - f_k d \quad \text{where } \Delta E_m = \Delta K + \Delta U$$

For a closed system with friction: $\Delta K + \Delta U = -f_k d$

P. 28) A 16 kg child descends a slide 2.20 m high and starting from rest, reaches the bottom with a speed of 1.15 m/s. How much thermal energy due to friction was generated in this process?

$$\begin{aligned} \text{Soln: } \Delta K + \Delta U_g &= -f_k d \quad (f_k d \text{ is the "thermal energy due to friction"}) \\ &\rightarrow f_k d = 334 \text{ J} \end{aligned}$$

P. 29) Ski starts from rest and slides down 28° incline 85 m long. a) if $\mu_k = 0.09$, what is the ski's speed at base of the incline?

$$\begin{aligned} \text{Soln: } \Delta K + \Delta U_g &= -f_k d \quad \text{Find } v_f \text{ buried in } \Delta K \\ v_i &= 0, mg\Delta y = mg85\sin 28 = mg(40\text{m}), d = 85 \text{ m}, f_k = 0.09mg\cos 28 \end{aligned}$$

See "Exam-like questions for Ch. 8", #9 and #10.

Power (Ch. 8)

(copied from chapter8.ppt)

Power is the time rate of energy change.

The ***instantaneous power*** is defined as

$$P \equiv \frac{dE}{dt}$$

Average power is defined as

$$P_{avg} \equiv \frac{\Delta E}{\Delta t}$$

- The SI unit of power is called the watt: 1 watt = 1 joule / second = 1 kg · m² / s³
- A unit of power in the US Customary system is horsepower: 1 hp = 746 W
- Units of power can also be used to express units of energy:
1 kWh = (1000 W)(3600 s) = 3.6 x 10⁶ J

The ΔE in P_{avg} could be an energy lost or gained, or an energy produced or consumed, and it could involve any type of energy: work (W), heat (Q), ΔU , ΔK , etc..

A common example in problems is: $P_{avg} = \frac{W}{\Delta t}$

Ex) (P. 56) How long will it take a 1750-W motor to lift a 335 kg piano to a sixth floor window 18.0 m above?

Soln: $P_{avg} = \Delta U_g / \Delta t$ with $\Delta U_g = mg\Delta y = (335)(9.8)(18.0) = 59090 \text{ J}$
(OR $P_{avg} = W_{mot} / \Delta t$ with $W_{mot} = mg\Delta y$.) So $\Delta t = mg\Delta y / P_{avg}$
 $\Delta t = (59090 \text{ J}) / (1750 \text{ J/s}) = 33.8 \text{ sec}$

Ex) (P. 57) An 85-kg football player traveling 5.0 m/s is stopped in 1.0 sec by a tackler. (a) What is the original kinetic energy of the player?
(b) What is the average power required to stop him?

Soln: (a) $K = \frac{1}{2} m v^2$. (b) $P_{avg} = \Delta K / \Delta t$

Power as force times velocity

Since differential work $dW = \mathbf{F} \cdot d\mathbf{r}$, the instantaneous power associated with work can be written:

$$P = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$

The $\mathbf{v}$ is the velocity of the point of application of the force doing the work.

Ex) (P. 58) If a car generates 18 hp when traveling at a steady 95 km/hr, what must be the average force exerted on the car due to friction and air resistance?

Soln: $P = Fv \rightarrow F = P/v$.

Convert $P = 18\text{hp}$ (746 W/hp) = 13428 W

$v = 95\text{ km/hr}$ (1000m/km)($1\text{hr}/3600\text{s}$) = 26.4 m/s. $F = 13428/26.4 = 509\text{ N}$

Linear Momentum

The **linear momentum** of an object of mass m moving with a velocity $\mathbf{v}$ is defined to be the product of the mass and velocity: $\mathbf{p} \equiv m\mathbf{v}$

- Linear momentum is a vector quantity.
- Its direction is the same as the direction of the velocity.

The dimensions of momentum are ML/T.

The SI units of momentum are kg · m / s.

Momentum can be expressed in component form:

$$\begin{array}{lll} \text{■ } p_x = m v_x & p_y = m v_y & p_z = m v_z \end{array}$$

Linear Momentum is also conserved in a closed system:

$$\mathbf{p}_{\text{tot}} = \mathbf{p}'_{\text{tot}}$$

Where $\mathbf{p}_{\text{tot}}$ is the total momentum before some interaction within the system, and $\mathbf{p}'_{\text{tot}}$ is the total momentum after the collision.

Why do we need momentum? Consider this problem...

A 60 kg archer stands on frictionless ice and fires a 0.5 kg arrow at 80 m/s. What is the archer's velocity after firing the arrow?

Approaches:

- Kinematics – no
 - No information about acceleration.
- Newton's Second Law – no
 - No information about F or a
- Energy conservation – no
 - No information about work or energy
- Momentum conservation – yes



© Cengage Learning. All Rights Reserved.

Why do we need momentum? Consider this problem...

A 60 kg archer stands on frictionless ice and fires a 0.5 kg arrow at 80 m/s. What is the archer's velocity after firing the arrow?

Momentum conservation:

$$\mathbf{p}_{\text{initial}} = m_{\text{arch}} \mathbf{v}_{\text{arch}} + m_{\text{arr}} \mathbf{v}_{\text{arr}} = 0 + 0 = 0$$

$$\mathbf{p}_{\text{final}} = m_{\text{arch}} \mathbf{v}_{\text{arch}} + m_{\text{arr}} \mathbf{v}_{\text{arr}} = (60\text{kg})\mathbf{v}_{\text{arch}} + (0.5\text{kg})(80\text{ m/s } \hat{\mathbf{i}})$$

$$\mathbf{p}_{\text{initial}} = \mathbf{p}_{\text{final}}$$

$$0 = (60\text{ kg}) \mathbf{v}_{\text{arch}} + (40\text{ kg m/s}) \hat{\mathbf{i}}$$

$$60\text{ kg } \mathbf{v}_{\text{arch}} = -40\text{ kg m/s } \hat{\mathbf{i}}$$

$$\mathbf{v}_{\text{arch}} = -0.67\text{ m/s } \hat{\mathbf{i}}$$

See



© Cengage Learning. All Rights Reserved.

Outline for W9,D3

Momentum, $\mathbf{p=mv}$

Relation to KE

Relation to force

Types of collisions

2-D Collisions

Homework

Ch. 9 P. 1,2,4-6,8,9,18,21,23,28,29,37,38,47,54,55 Due Mon

Notes: “NEW STUFF”:

Practice quizzes, exam-like Qs, Ch.8 PPT, Ch 9 PPT,
old notes with examples, etc.

Momentum and Kinetic Energy

Momentum and kinetic energy both involve mass and velocity.

There are major differences between them:

- Kinetic energy is a scalar and momentum is a vector.
- Kinetic energy is proportional to v^2 , momentum is proportional to v .

Ex) If $m_1 = 4m_2$ and $K_1 = K_2$, what is p_1/p_2 ?

(p_1 is the magnitude of $\mathbf{p}_1$)

Ans: $\frac{1}{2} m_1 v_1^2 = \frac{1}{2} m_2 v_2^2$ so

$$4m_2 v_1^2 = m_2 v_2^2$$

$$v_1^2/v_2^2 = 1/4$$

$$v_1/v_2 = 1/2$$

$$\text{So } p_1/p_2 = m_1 v_1 / m_2 v_2$$

$$p_1/p_2 = (4)(1/2) \\ = 2$$

The more massive object (m_1) has smaller v (if K 's are equal).

The more massive object (m_1) has larger p (if K 's are equal).

The rate of change of Momentum is *Force*

Newton's Second Law can be used to relate the momentum of a particle to the resultant force acting on it.

$$\vec{\Sigma \mathbf{F}} = m\vec{\mathbf{a}} = m \frac{d\vec{\mathbf{v}}}{dt} = \frac{d(m\vec{\mathbf{v}})}{dt} = \frac{d\vec{\mathbf{p}}}{dt}$$

with constant mass.

Thus, $\mathbf{F}_{\text{net}} = d\mathbf{p}/dt$

(The time rate of change of the linear momentum of a particle is equal to the net force acting on the particle.)

- This is the form in which Newton presented the Second Law.
- It is a more general form than the one we used previously.
- This form also allows for mass changes.

Integral of Force gives change in momentum

Change of momentum is called impulse.

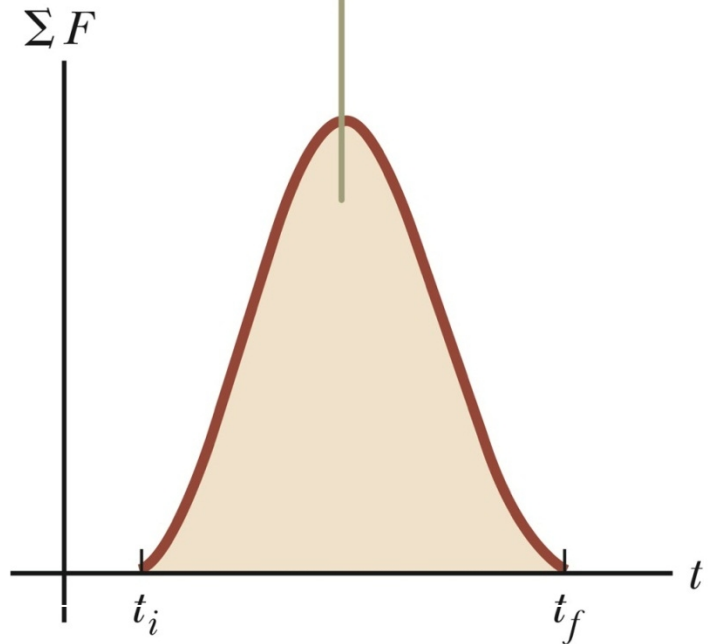
$$\mathbf{I} = \Delta \mathbf{p}$$

The magnitude of the impulse is equal to the area under the force-time curve.

- The force may vary with time.

Thus,
$$\Delta \vec{p} = \int_{t_1}^{t_2} \vec{F}_{net} dt$$

The impulse imparted to the particle by the force is the area under the curve.



a

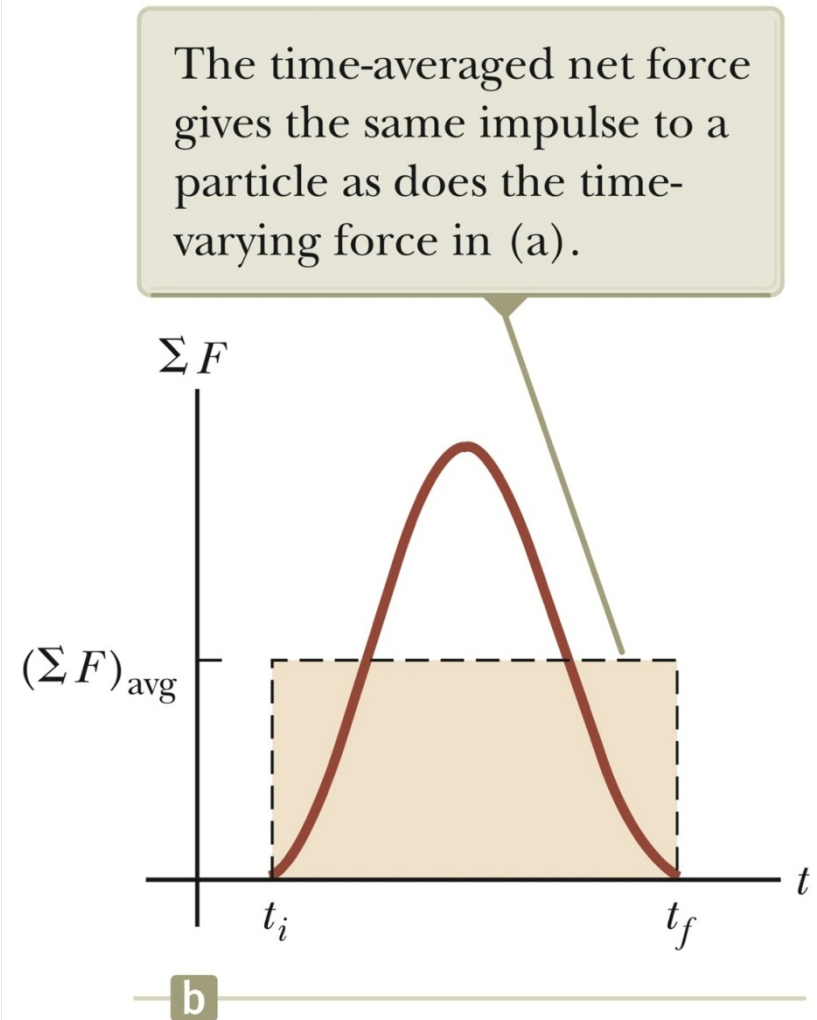
Change of momentum or Impulse

The impulse can also be found by using the time averaged net force.

$$\mathbf{I} = \sum \mathbf{F} \Delta t$$

This would give the same impulse as the time-varying force does.

$$\Delta p = \vec{F}_{net, avg} \Delta t$$



© Cengage Learning. All Rights Reserved.

Examples of the relations between $\mathbf{F}$ and $\mathbf{p}$

Ex) (P. 5) Find $\mathbf{F}(t)$ if $\mathbf{p} = 4.8t^2 \hat{i} - 8 \hat{j} - 9.4t \hat{k}$.

Ex) (P. 8) Find $\Delta \mathbf{p}$ between $t=1.0$ and $t=2.0$ sec if $\mathbf{F}=26\hat{i}-12t^2\hat{j}$.

Derive conservation of momentum from Newton's 3rd law!

Consider a collision between 2 compressible balls, m_1 and m_2 .

[See slow-motion videos of ball collisions.]

During every instant of contact, $\mathbf{F}_{1\text{by}2} = -\mathbf{F}_{2\text{by}1}$ (by Newton's 3rd).

So $m_1 d\mathbf{v}_1/dt = -m_2 d\mathbf{v}_2/dt$ But if m 's are constant, this means

$$d(m_1 \mathbf{v}_1)/dt = -d(m_2 \mathbf{v}_2)/dt$$

Or $d\mathbf{p}_1/dt + d\mathbf{p}_2/dt = 0$ which is $d(\mathbf{p}_1 + \mathbf{p}_2)/dt = 0$ or $\Delta \mathbf{p}_{\text{tot}} = 0$

Conservation of Linear Momentum

Whenever two or more particles in an isolated system interact, the total momentum of the system remains constant.

For a 2-body system:
$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = m_1 \vec{v}'_1 + m_2 \vec{v}'_2$$

For an N-body system:
$$\sum_{i=1}^N m_i \vec{v}_i = \sum_{i=1}^N m_i \vec{v}'_i$$

Or:
$$\vec{p}_{tot} = \vec{p}'_{tot}$$

For a 2-body 1D collision:
$$m_1 v_{1x} + m_2 v_{2x} = m_1 v'_{1x} + m_2 v'_{2x}$$

For a 2-body 2D collision, include this equation with the above:

$$m_1 v_{1y} + m_2 v_{2y} = m_1 v'_{1y} + m_2 v'_{2y}$$

Conservation of Linear Momentum

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = m_1 \vec{v}'_1 + m_2 \vec{v}'_2$$

Example: (P. 6) A child in a stationary boat throws a 5.3 kg package out the back with a speed of 10 m/s. Calculate the velocity of boat immediately after, assuming it was initially at rest. The mass of the child is 24 kg and the mass of the boat is 42 kg.

Soln: Apply cons of momentum to a 2-mass system.

Let $m_1 = m_{\text{child}} + m_{\text{boat}} = 66 \text{ kg}$ (the child is fixed w.r.to the boat)
and let $m_2 = m_{\text{package}} = 5.3 \text{ kg}$

So
becomes

$$\begin{aligned} m_1 \vec{v}_1 + m_2 \vec{v}_2 &= m_1 \vec{v}'_1 + m_2 \vec{v}'_2 \\ 0 + 0 &= 66 \mathbf{v}'_1 + 5.3 \mathbf{v}'_2 \\ 0 &= 66 \mathbf{v}'_1 + 5.3 (10 \text{ m/s} \hat{i}) \\ -53 \text{ kg m/s } \hat{i} &= 66 \text{ kg } \mathbf{v}'_1 \\ -0.8 \text{ m/s } \hat{i} &= \mathbf{v}'_1 \end{aligned}$$

Example: see “Scan of old Ch. 9 notes”, 2nd page, for girl on plank problem.

3 Types of Collisions

For ALL 3 types of collisions, momentum is conserved!!

Elastic collision: kinetic energy is also conserved.

- Elastic collisions occur on a microscopic level between molecules.
- In large-scale collisions, only approximately elastic collisions actually occur.

- Special equations for the 1-D, head-on case: $(v_A - v_B) = -(v'_A - v'_B)$

$$v'_B = v_A \left(\frac{2m_A}{m_A + m_B} \right) + v_B \left(\frac{m_B - m_A}{m_A + m_B} \right) \quad v'_A = v_A \left(\frac{m_A - m_B}{m_A + m_B} \right) + v_B \left(\frac{2m_B}{m_A + m_B} \right)$$

- See Prob. 34 in Giancoli. DEMO: **9.11.swf head-on collisions**

Inelastic collision: kinetic energy is *not* conserved.

- Most real-life collisions.

Perfectly inelastic collision: kinetic energy is lost AND the objects stick together

- Special equation for 2-body case: $m_1 \vec{v}_1 + m_2 \vec{v}_2 = (m_1 + m_2) \vec{v}'$

Elastic Collisions, cont.

Example of some special cases of elastic collisions:

- $m_1 = m_2$ The particles exchange velocities
- When a very heavy particle collides head-on with a very light one initially at rest, the heavy particle continues in motion unaltered and the light particle rebounds with a speed of about twice the initial speed of the heavy particle.
- When a very light particle collides head-on with a very heavy particle initially at rest, the light particle has its velocity reversed and the heavy particle remains approximately at rest.

Demonstrate head-on collisions with different mass ratios: 9.11.swf

DEMO: “supernova core collapse and rebound”

Demonstrate glancing collisions with equal mass ratios: 9.11.swf

What is the angle between the final velocities?

Demonstrate glancing collisions with arbitrary masses: 9.11.swf

Save for 2-D collisions.

Elastic Collisions, cont.

Prove that the final height of the small ball in the “supernova core collapse” demo is a maximum of 4 times its starting height.

Part 1. When m_1 collides elastically and head-on with a stationary m_2 , show that v_2' / v_1 approaches 2.0 as m_1 / m_2 approaches infinity (∞).

DEMO: Use 9.11.swf: $m_1=20$, $m_2=5$ ($m_1/m_2=4$), $v_1=10$, then $v_{2f}/v_{1f}=16/10=1.6$.

Use this eqn for elastic collisions:

$$v_B' = v_A \left(\frac{2m_A}{m_A + m_B} \right) + v_B \left(\frac{m_B - m_A}{m_A + m_B} \right)$$
$$v_2' = v_1 \left(\frac{2m_1}{m_1 + m_2} \right) + 0 \quad \text{or} \quad \frac{v_2'}{v_1} = \lim_{m_1 \gg m_2} \left(\frac{2m_1}{m_1 + m_2} \right) = 2.0$$

Part 2. Find max height y'' above table for m_2 when it falls on top of m_1 after falling a distance y .

Find rebound speed of m_1 : $v_1'^2 - 0^2 = 2(-9.8)(0 - y)$ gives $v_1' = (19.6y)^{1/2}$

Find rebound speed of m_2 : $v_2' = 2.0 v_1' = 2.0 (19.6y)^{1/2}$ (from Part 1)

Find y'' using: $v_2''^2 - v_2'^2 = 2a(y_2'' - y_2')$ (with $y_2' = 0$)

$$0 - 4.0 (19.6y) = -19.6(y_2'') \quad \text{or} \quad y_2''/y = 4.0$$

Outline for W10,D1

Elastic collisions – extreme m_1/m_2 cases

Perfectly inelastic collision – ballistic pendulum

2D-collision example

Demo: Newton's Cradle

Center of mass

Homework

Ch. 9 P. 1,2,4-6,8,9,18,21,23,28,29,37,38,47,54,55 Due 2 pm

Skip 9.10, read 9.1-9.9.

Ch. 10 P. 1,4-6,19-21,25,28-30,34,35,37,53,54,55,64,67,69

Do for next Wed

Notes: NEW STUFF: I've added ppt outlines for future chapters.

This weeks lab: 2D conservation of momentum

Next exam after Ch. 11.

Perfectly Inelastic Collision Example – Ballistic Pendulum

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = (m_1 + m_2) \vec{v}'$$

Conceptualize

- The bullet enters the pendulum, and gets embedded before the block rises measurably.
- Then block+bullet swings up a height h before coming to a stop.

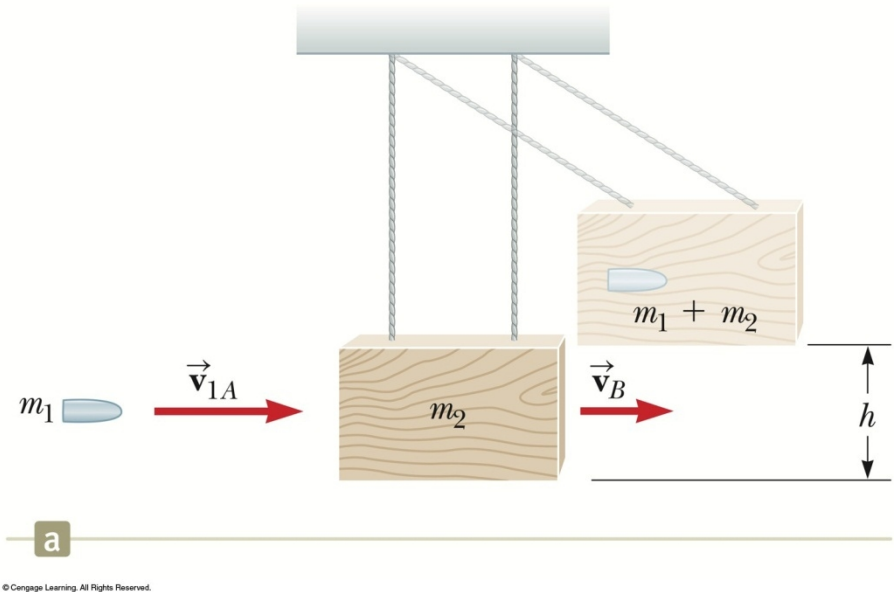
We can use conservation of $\mathbf{p}$ and conservation of E_{mech} to find the speed of the bullet.

Consider 2 time intervals:

$\Delta T1$: m_b in motion at v_b to m_b stuck in wood m_w moving at v' .

$\Delta T2$: wood at $h=0$ with v' to wood at $h=h$ with $v_f=0$.

Use $\Delta T1$ for $m_b v_b + m_w v_w = (m_b + m_w) v'$

$$v' = \left(\frac{m_b}{m_b + m_w} \right) v_b$$


Use $\Delta T2$ for $\Delta K = -\Delta U_g$

$$\frac{1}{2}(m_w + m_b)(0 - v'^2) = -mgh$$

$$v' = (2gh)^{1/2}$$

Combine:

$$v_b = \left(\frac{m_b + m_w}{m_b} \right) \sqrt{2gh}$$

Ballistic Pendulum, cont.

A multi-flash photograph of a ballistic pendulum.

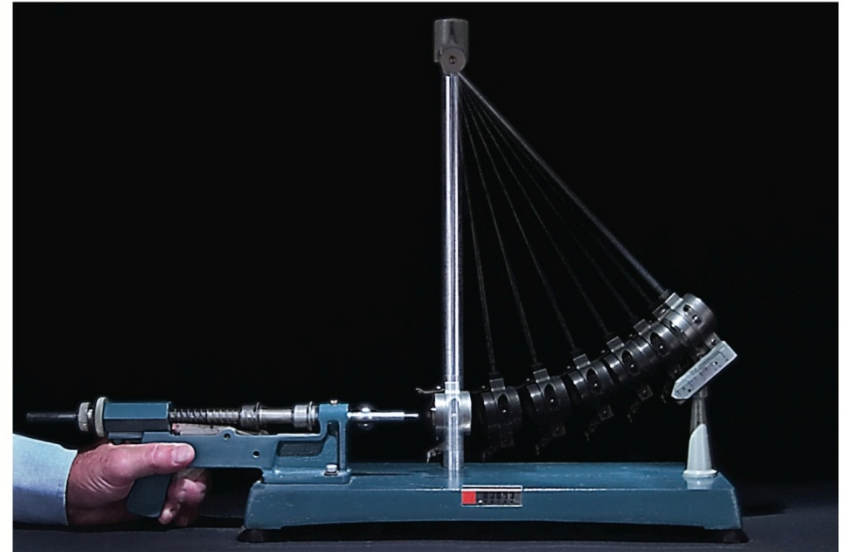
Ex) Find v_b if $m_b = 0.05$ kg, $m_w = 4$ kg and $h = 7$ cm.

$$v_b = \left(\frac{m_b + m_w}{m_b} \right) \sqrt{2gh}$$

$$V_b = (.05 + 4) / (.05) (2(9.8)(.07))^{1/2}$$

$$V_b = 93.9 \text{ m/s}$$

© Cengage Learning/Charles D. Winters



b

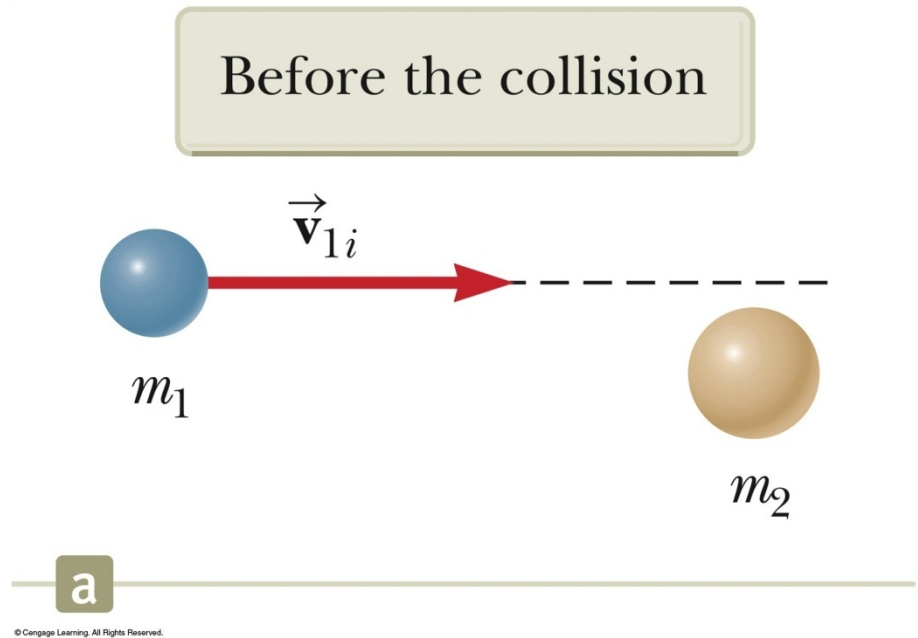
Two-Dimensional Collision – the glancing, 2-body elastic collision

The momentum is conserved independently in x-, y-, and z-components.

Mass m_1 is moving at velocity $\mathbf{v}_{1i}$ and mass m_2 is at rest.

In the x-direction, the initial momentum is $p_x = m_1 v_{1i}$

In the y-direction, the initial momentum is $p_y = 0$.



Two-Dimensional Collision, example cont.

After the collision, the momentum in the x-direction is

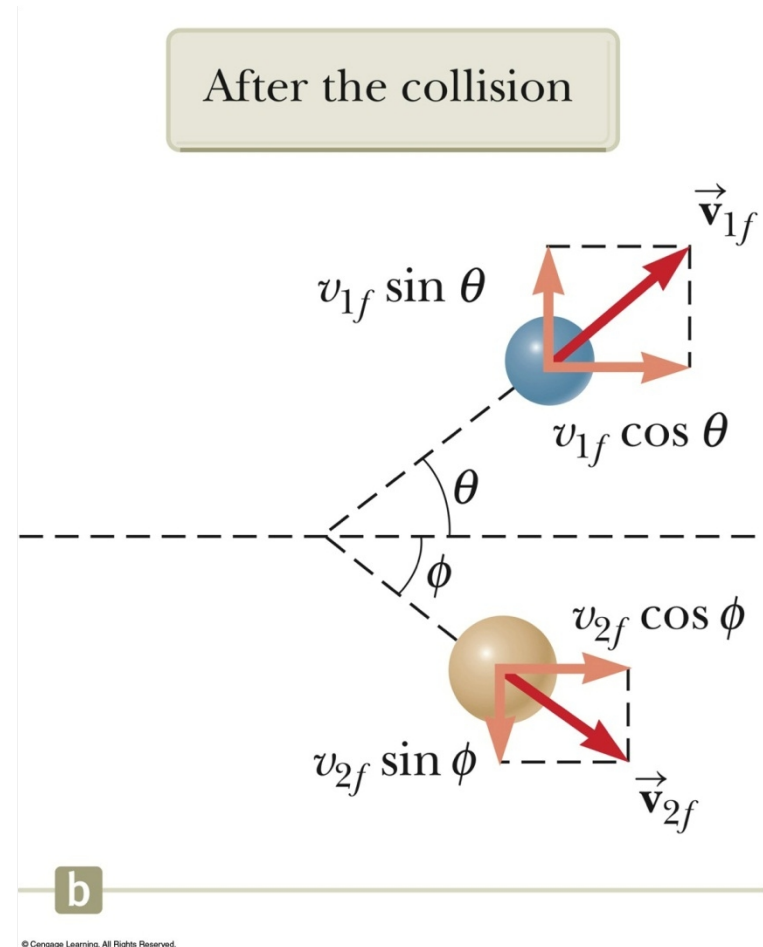
$$p_x' = m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \phi.$$

After the collision, the momentum in the y-direction is

$$p_y' = m_1 v_{1f} \sin \theta - m_2 v_{2f} \sin \phi.$$

- The negative sign is due to the component of the velocity being downward.

If the collision is elastic, you can also use a kinetic energy equation.



Two-Dimensional Collisions

Before:

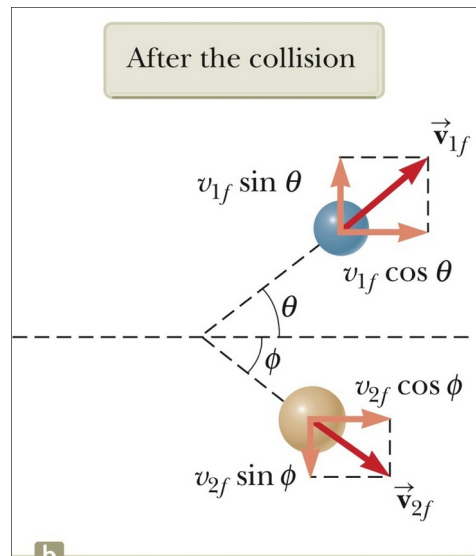
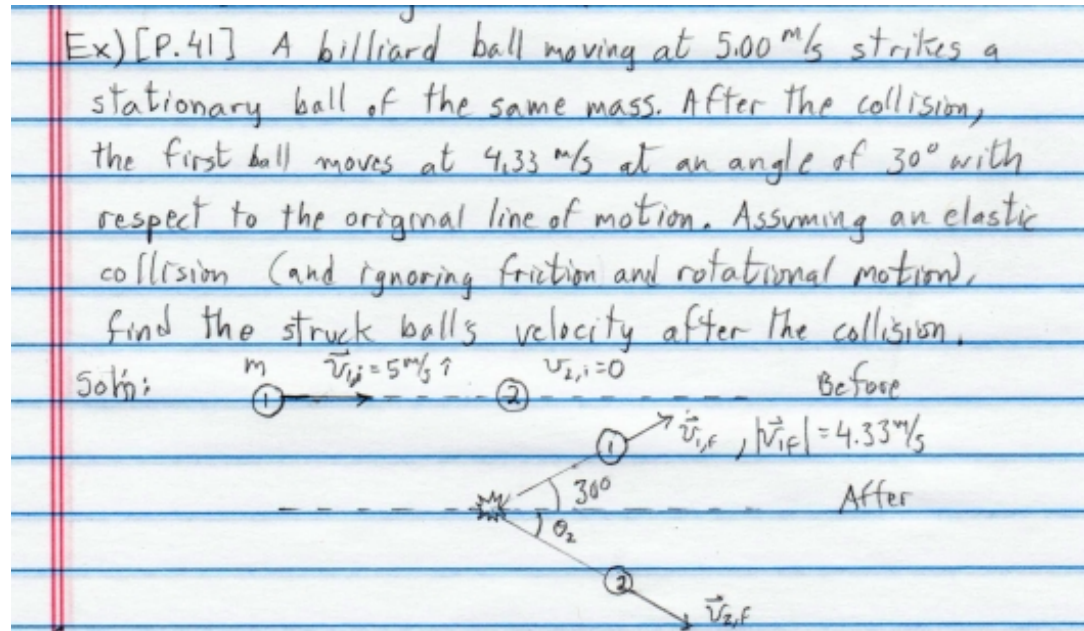
$$p_x = m_1 v_{1i}, \quad p_y = 0$$

After the collision:

$$p'_x = m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \phi$$

$$p'_y = m_1 v_{1f} \sin \theta - m_2 v_{2f} \sin \phi$$

See "ExampleCh9p2.jpg"



$$p_x = p'_x$$

$$m(5.0) = m(4.33)\cos(30) + m v_{2x}'$$

$$v_{2x}' = 5.0 - (4.33)\cos(30) = 1.25 \text{ m/s} = v_2' \cos(\phi)$$

$$p_y = p'_y$$

$$0 = m(4.33)\sin(30) + m v_{2y}'$$

$$v_{2y}' = -4.33\sin(30) = -2.165 \text{ m/s} = v_2' \sin(\phi)$$

$$\text{So } \mathbf{v}_2' = 1.3\hat{i} - 2.2\hat{j} \text{ m/s or } 2.50 \text{ m/s at } \phi = -60^\circ$$

Example: Executive Time Waster (Newton's Cradle)

What happens when 1 drops?

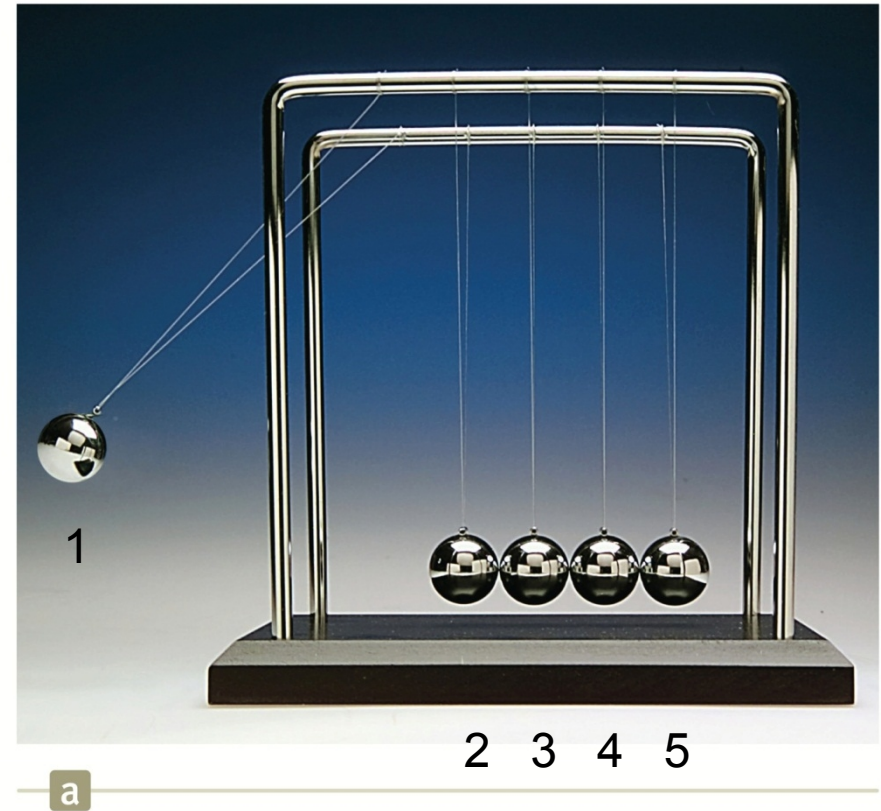
What happens when 1+2 drop?

How about 1+2+3?

What do these all have in common?

Do the 5 ball bearings comprise a closed system?

How about the balls + supports?



© Cengage Learning/Charles D. Winters

Executive Time Waster (cont.)

Why can't ball 1 fall and 4+5 rise up?

- momentum can be conserved!
- but can energy?

What if you fuse balls 4 & 5 together?

- calculate v 's with $m_A = m$, $m_B = m_4 + m_5 = 2m$

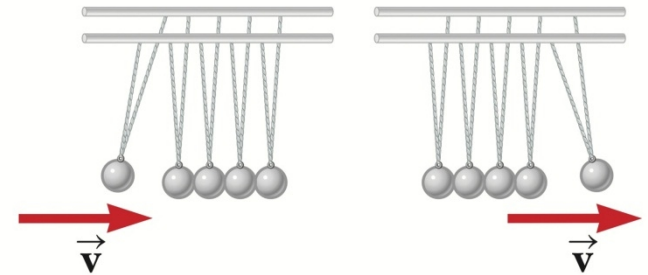
$$v'_B = v_A \left(\frac{2m_A}{m_A + m_B} \right) + v_B \left(\frac{m_B - m_A}{m_A + m_B} \right)$$

$$v'_A = v_A \left(\frac{m_A - m_B}{m_A + m_B} \right) + v_B \left(\frac{2m_B}{m_A + m_B} \right)$$

You should get:

$$v'_B = 0.66 v_A \text{ and } v'_A = -0.33 v_A$$

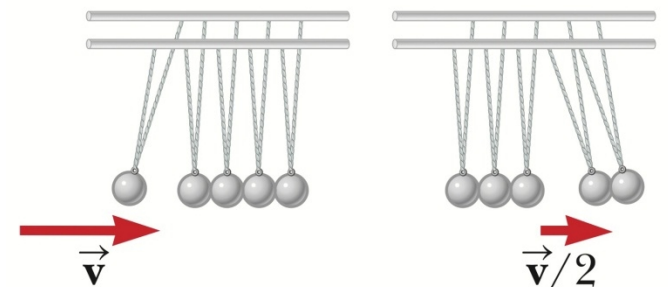
This can happen



b

© Cengage Learning. All Rights Reserved.

This cannot happen



c

© Cengage Learning. All Rights Reserved.

The Center of Mass

The **center of mass**, COM, is the balance point of a system.

- If the system is a single, extended object, you can put your finger under the COM to balance the object. (*Examples: a ruler, a boomerang*)
- If the system is isolated, its momentum is conserved, and the COM moves at a constant velocity. (DEMO: 9.13.swf)
- *If the system is a collection of particles, the COM is the mass-weighted average position of all particles.*

Here's an average position:

$$x_{avg} = \frac{\sum_i x_i}{N}$$

Here's a mass-weighted average position:

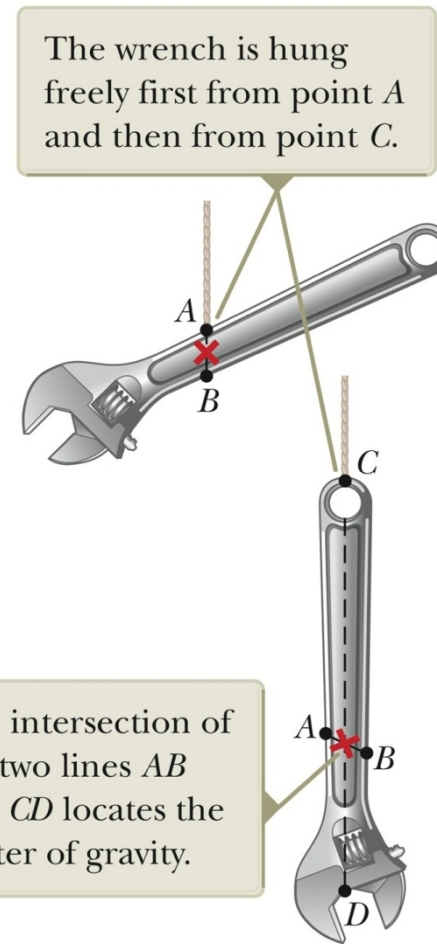
$$x_{avg} = \frac{\sum_i m_i x_i}{\sum_i m_i} = \frac{\sum_i m_i x_i}{M}$$

Finding Center of Gravity, Irregularly Shaped Object

Suspend the object from one point.

Then, suspend from another point.

The intersection of the resulting lines is the center of mass / gravity and half way through the thickness of the wrench.



© Cengage Learning. All Rights Reserved.

Center of Mass, Coordinates

The coordinates of the center of mass are

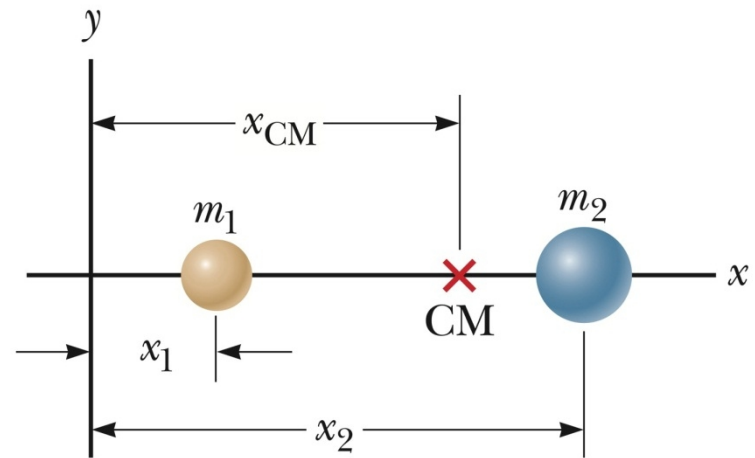
$$x_{\text{CM}} = \frac{\sum_i m_i x_i}{M} \quad y_{\text{CM}} = \frac{\sum_i m_i y_i}{M}$$
$$z_{\text{CM}} = \frac{\sum_i m_i z_i}{M}$$

- M is the total mass of the system.
 - See 9.14.swf
 - DEMO: masses on a ruler. Try the following:

0.5 kg at 30 cm and 1.0 kg at 60 cm. $X_{\text{avg}} = ?$ $x_{\text{com}} = ?$

0.5 kg at 0 cm and 1.0 kg at 30 cm. $X_{\text{avg}} = ?$ $X_{\text{com}} = ?$

0.5 kg at 0 cm, 1.0 kg at 30 cm, and $M_{\text{ruler}} = \underline{\hspace{1cm}}$ at 50 cm. $X_{\text{com}} = ?$



© Cengage Learning. All Rights Reserved.

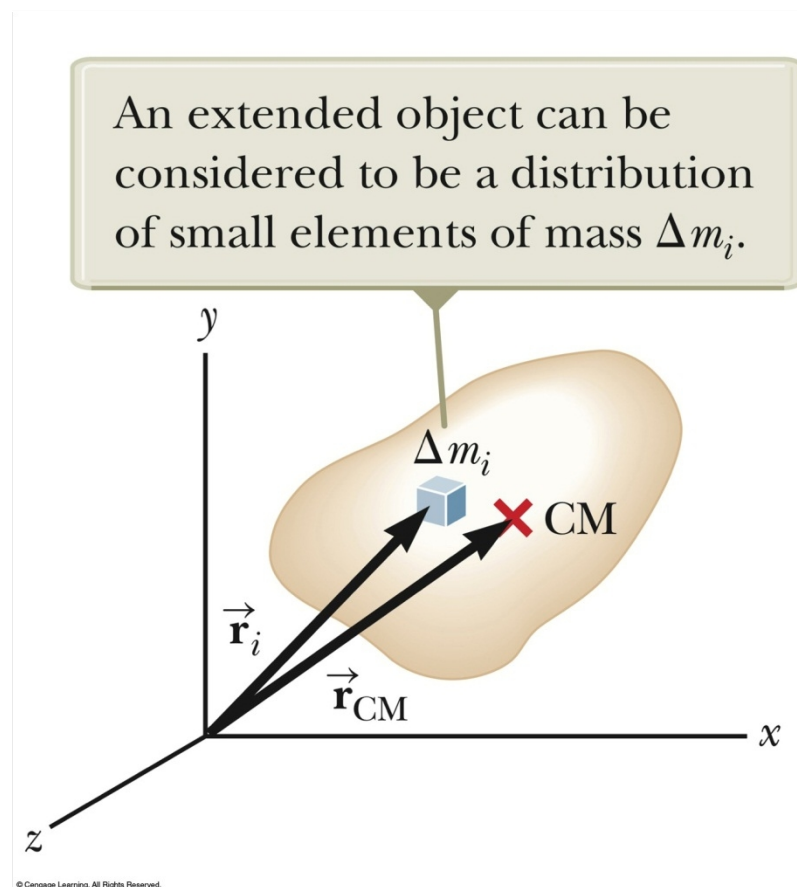
Center of Mass, Extended Object

Similar analysis can be done for an extended object.

Consider the extended object as a system containing a large number of small mass elements.

As the elements, Δm , become tiny, it can be considered to have a continuous mass distribution.

$$\vec{r}_{\text{CM}} = \frac{1}{M} \int \vec{r} dm$$



Center of Mass, position

The center of mass in three dimensions can be located by its position vector, $\vec{r}_{CM}$.

- For a system of particles,

$$\vec{r}_{CM} = \frac{1}{M} \sum_i m_i \vec{r}_i$$

- $\vec{r}_i$ is the position of the i th particle, defined by

$$\vec{r}_i = x_i \hat{i} + y_i \hat{j} + z_i \hat{k}$$

- So, for an extended object, $\vec{r}_{CM} = \frac{1}{M} \int \vec{r} dm$

- dm can be λdx , σdA , or ρdV
- Where e.g., λ = mass/length (kg/m), is a linear mass density.
- And the integral is over the space occupied by the mass.
- For non-uniform compositions, λ can be a f'n of x , like $\lambda(x) = ax + b$

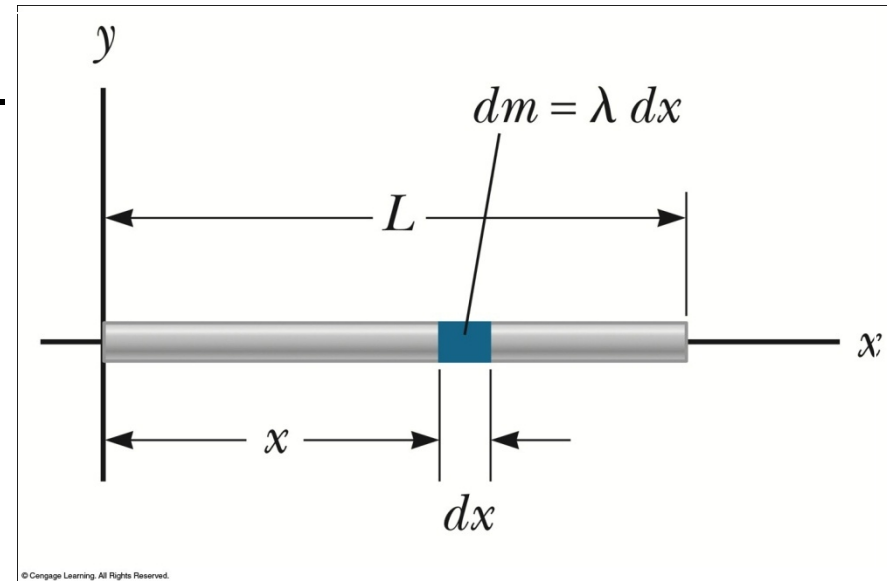
Ex) Show that the COM of a uniform rod of length L and mass M is a distance $L/2$ from the end.

- Imagine the rod aligned with the x -axis, with one end on $(0,0)$.
- The COM will have $y_{\text{COM}} = z_{\text{COM}} = 0$.

Do integral using $\lambda = \text{constant}$

$$\begin{aligned}x_{\text{COM}} &= \frac{1}{M} \int_0^L x \lambda dx \\x_{\text{COM}} &= \frac{1}{M} \int_0^L x \lambda_0 dx \\x_{\text{COM}} &= \frac{\lambda_0}{M} \left(\frac{L^2}{2} - 0 \right)\end{aligned}$$

But $\lambda_0 = M/L$, so $x_{\text{COM}} = 1/L(L^2/2) = L/2$ Q.E.D



Ex) Find the COM of a non-uniform rod of length 1.0 m if its linear mass distribution is $\lambda(x)=3x+1$ kg/m, where $x=0$ at the origin.

- As before, rod is aligned with the x -axis, with one end on $(0,0)$, and $y_{\text{COM}} = z_{\text{COM}} = 0$.

Do integral using $\lambda = 3x+1$

$$x_{\text{COM}} = \frac{1}{M} \int_0^L x \lambda dx$$

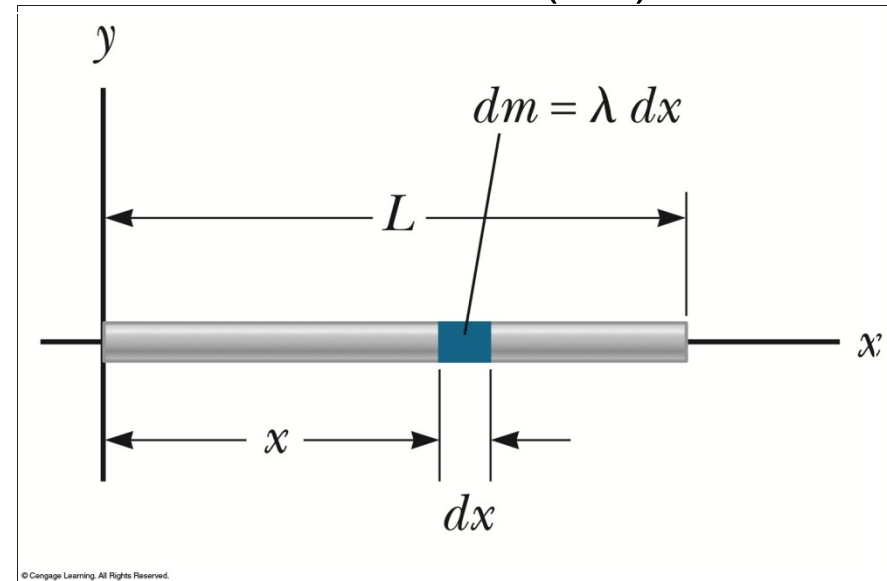
$$x_{\text{COM}} = \frac{1}{M} \int_0^{1.0} x (3x+1) dx$$

$$x_{\text{COM}} = \frac{1}{M} \left(x^3 + \frac{x^2}{2} \right)_0^{1.0}$$

So $x_{\text{com}} = 1.5/M$, but what is M ? M is the total mass.

$$M = \int_0^{1.0} \lambda dx = \int_0^{1.0} (3x+1) dx = \left(\frac{3x^2}{2} + x \right)_0^{1.0} = 5/2.$$

$$\text{So } x_{\text{com}} = (3/2)/(5/2) = 3/5 = 0.6 \text{ m}$$



Velocity and Momentum of a System of Particles

The velocity of the center of mass of a system of particles is

$$\vec{\mathbf{v}}_{\text{CM}} = \frac{d\mathbf{r}_{\text{CM}}}{dt} = \frac{1}{M} \sum_i m_i \vec{\mathbf{v}}_i \quad (\text{it's a mass-weighted velocity!})$$

The momentum can be expressed as

$$M \mathbf{v}_{\text{CM}} = \sum_i m_i \mathbf{v}_i = \sum_i \mathbf{p}_i = \mathbf{p}_{\text{tot}}$$

The total linear momentum of the system, $\mathbf{p}_{\text{tot}}$, is the same as the momentum of a mass M (the total mass) multiplied by the velocity of the center of mass, $\mathbf{v}_{\text{CM}}$.

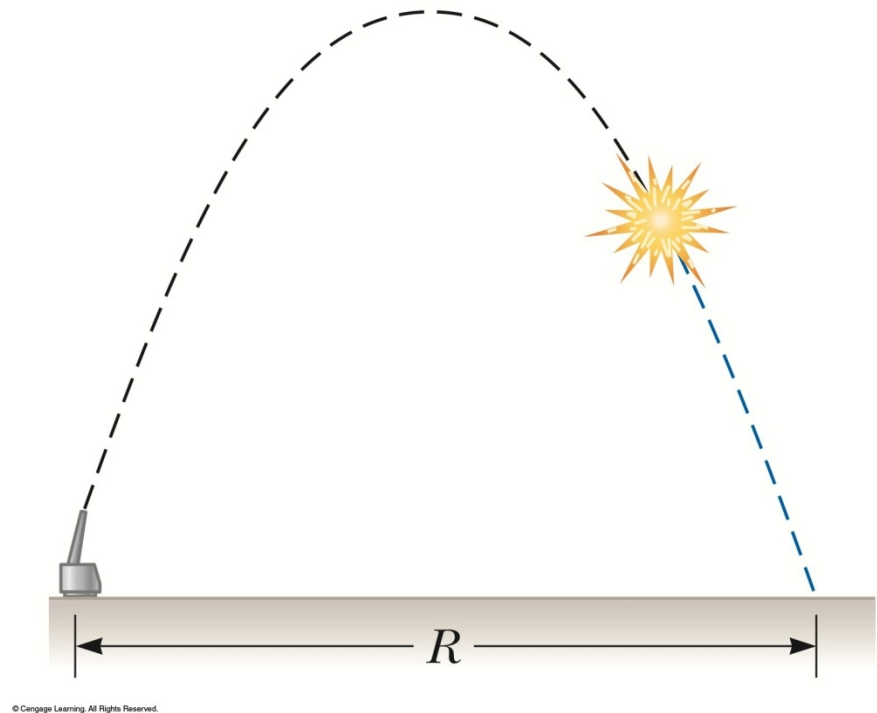
Since $\mathbf{p}_{\text{tot}}$ is conserved for an isolated system, the center of mass of the system must travel in a straight line (or in a parabolic trajectory if the only outside force is the force of gravity).

Motion of the Center of Mass, Example

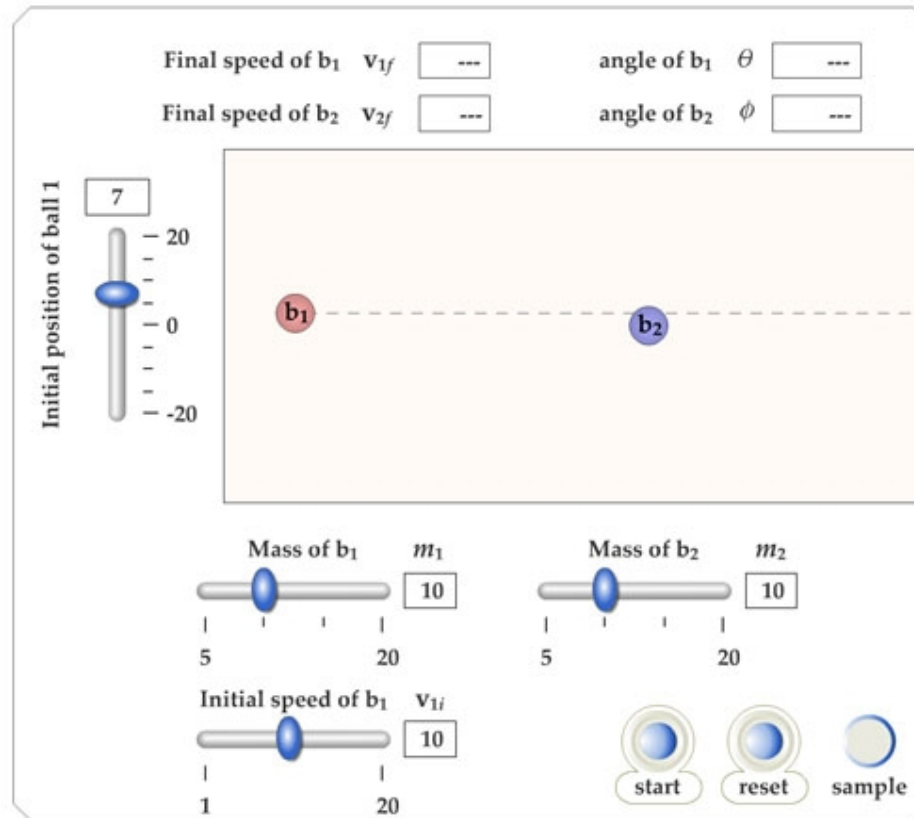
A projectile is fired into the air and suddenly explodes.

With no explosion, the projectile would follow the dotted line.

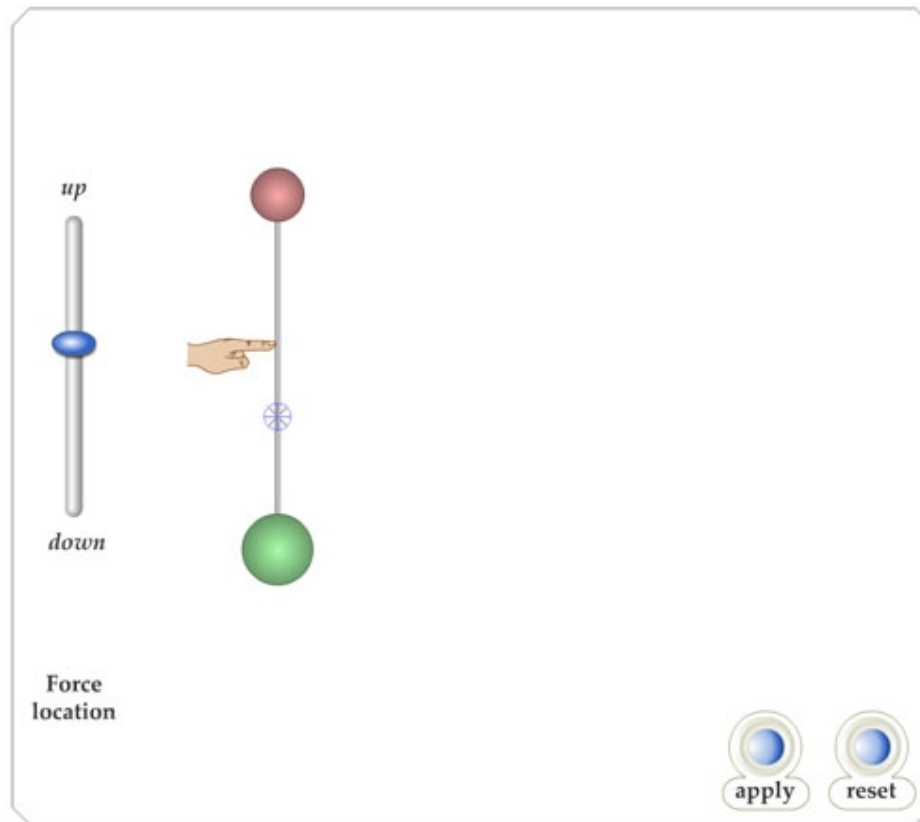
After the explosion, the center of mass of the fragments still follows the dotted line, the same parabolic path the projectile would have followed with no explosion.



9.11 Collisions in Two Dimensions



9.13 The Center of Mass



9.14 Calculating the Center of Mass

